



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc. (General) Degree in Information and Communication Technology
First Year - Semester I Examination
Modal Paper
ICT 1404 – Mathematics and Statistics for Computing

Time: Three (03) hours

Instructions

- **Answer any FIVE (05) questions.**
 - Calculators are allowed.
 - Each question carries equal marks.
 - Statistical tables and formulas are provided as attachments.
-

1. a) **I** How many different words can be made using all letters of the word 'MISSISSIPPI?
II In how many of the distinct permutations of the letters in MISSISSIPPI do the four same letter not come together?

(08 Marks)
- b) In a AGA division, there are 52 families, of which 22 families have at least 2 children. In a rural development programme, 20 families are to be chosen for fund assistance, of which at least 18 families must have at least 2 children. In how many ways can the choice be made?

(12 Marks)
2. a) If $A = \begin{pmatrix} 1 & 2 \\ 5 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -2 \\ 3 & 5 \end{pmatrix}$.
Find (i) $A+B$ (ii) AB (iii) A^{-1}

(06 Marks)
- b) Using matrices or Cramer's rule solve the following linear equations.
 $2x+3y+3z=15$
 $7x+6y+5z=38$
 $9x+10y+9z=56$

(14 Marks)
3. a) A basket contains 4 red, 5 blue and 3 green marbles. If 2 marbles are drawn at random from the basket, what is the probability that both are red?

(08 Marks)

- b) An student travels to college by three routes A, B and C. The probabilities he choose those routes are respectively 0.6, 0.3 and 0.1. The probabilities that he is late for college if he go via route A, B and C respectively are 0.02, 0.04 and 0.05. Find the probability he is late for college.

(12 Marks)

4. a) At a super market, 90% of customers pay by bank cards. Find the probability that in a randomly selected sample of five customers,
- Exactly two customers pay by bank cards,
 - More than two customers pay by bank cards.

(10 Marks)

- b) On average a photo copy machine breaks down four times during a month. Assuming that the number of breaks down can be modeled by a Poisson distribution, find the probability that it breaks down
- Exactly once in a given month,
 - At most two times in a 2 week duration.

(10 Marks)

5. The weight of a sugar packet which is normally distributed with a mean 500g and standard deviation 20g. Find the probability that the weight of a randomly selected sugar packet is
- less than 480g
 - more than 510g.
 - between 480g and 515g

What is the highest weight of the sugar packet out of the 10% percent of sugar packet which are overweight ?

(20 Marks)

6. The information of age and value of a car is given by the following table.

Age, x (years)	Value, Rs. (million)
2.0	6.1
2.5	5.5
3.0	5.0
3.5	4.6
4.5	3.8
5.0	3.5
6.0	3.3
7.0	1.0

- a) Calculate the equation of line of regression of y on x using least square method. **(10 Marks)**
- b) Estimate the value of the car when the age is 4 years. **(04 Marks)**
- c) Compute the coefficient of Determination. **(06 Marks)**

--END--